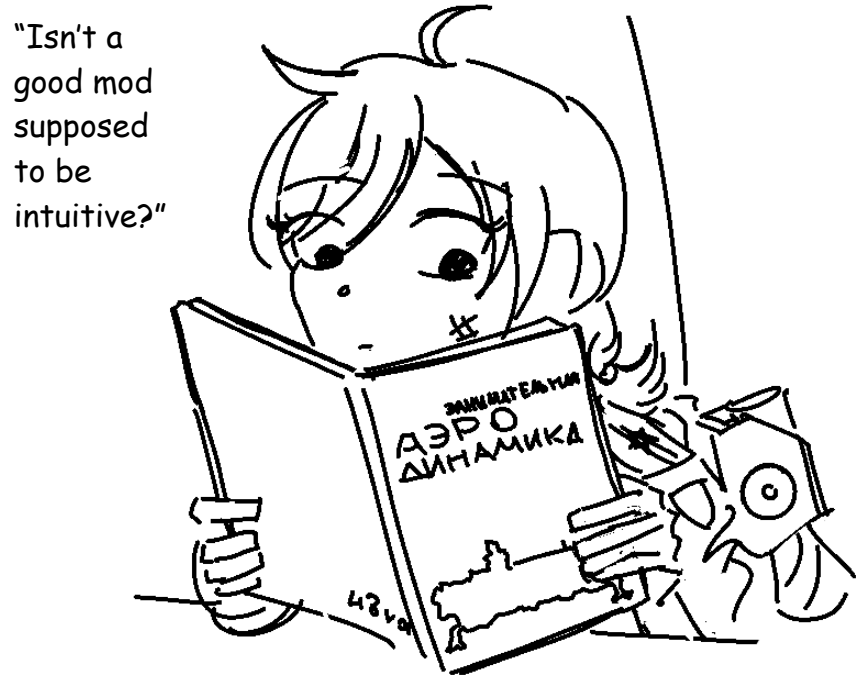
HIGHFLEET-ALLIES + AI MOD

Introduction and tuning guide for tarkhans of all age and descent.

INTRO

Welcome to Highfleet Allies.



Whether you're a seasoned player or a newbie who wants a different flavor of Highfleet, I urge you read this manual first because this mod adds a lot.

A lot, as follows:

- Updated AI subroutines for tac combat. Ships can now:
  - evade missiles
  - use flares
  - fire more than one Zenith at once
  - sidestep your shots
  - pull away when you're doing an underpass to unload a volley
- Ally system restored and brought back.
  - You can bring several allies in combat.
  - Allies inherit the same AI improvements as enemies.
  - Allies can also be used to bring carriers and provide CAS in combat
  - You can cycle between allies, or switch between two displayed ones

- Right, they are also displayed next to your ship panel
- Allies bomb landed targets
- Allies ignore prize ships
- Allies retreat on command
- Allies retreat automatically when their fuel runs low

- Carrier loadouts
  - Carriers can now equip ANY plane loadout in tac combat. Vanilla game already supported equipping 122mm rockets, but now you can equip anything.
  - T-7 with K-13 turn into interceptors.
  - You can put bombs on your La-29 and throw them in tac combat in a ditch.
  - Supposed to work with custom planes too. Not hardcoded and reads info from QoL mod JSON config
- Debug/Mod tools
  - This mod comes equipped with a massive optional logging tool that I called “megalog”. It can track crashes with more details, it can track GUI elements, every creature that appears on screen, so on. It is also the reason my logs folder is 3gb after a month of testing this mod.
  - A lot of mod makers will hopefully find debug overlay useful.
    It draws circles at fixed units from every ship, displays their AI state and waypoint. In the future I plan to expand it.



- Misc
  - QoL mod can now adjust refuel rate, change speed formula (in case you missed 1.151) and elevation rules
  - Fixed 2-3 crashes that were part of vanilla game

ALLY SYSTEMS

Most of the mod is actually quite intuitive



Ship in slot 1 is ALWAYS player ship.

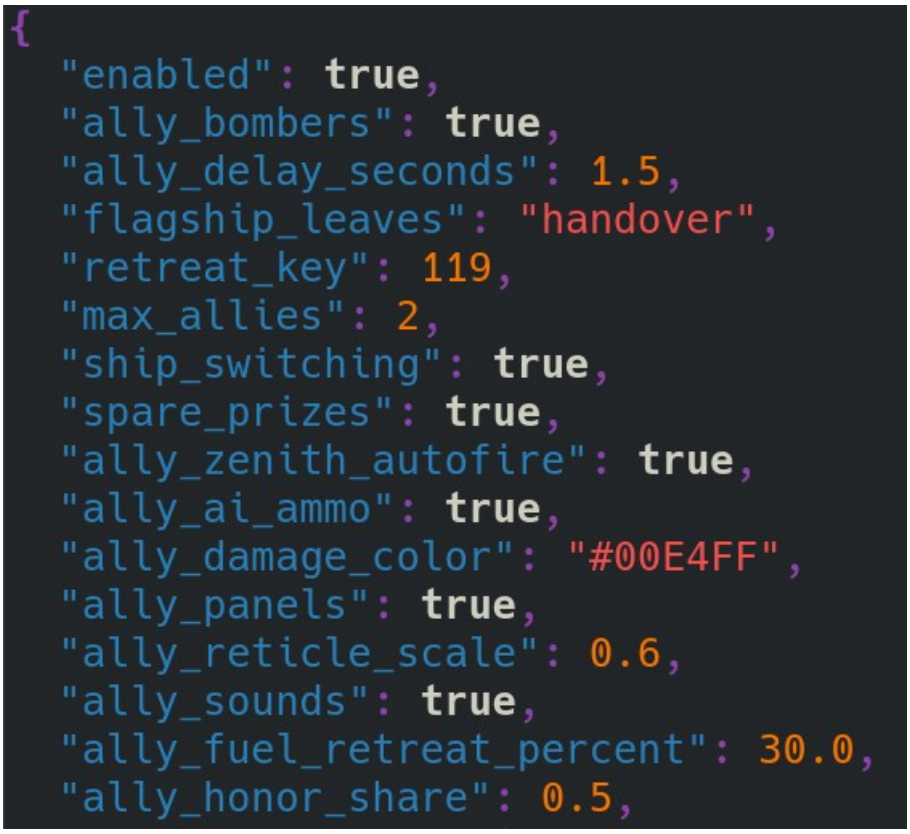
Every other ship can be assigned either a default role (so just like in vanilla it will only show up once you retreat), or it can be set to become a support ship.

Buttons are bound to slots, not to ships, do pay attention to it.

TUNING THE ALLY SYSTEM

Head to your game's folder, then Modloader – config – **ally.json.** Open it with any text editor, even Notepad will do.

Now, let's go through the mod settings.



This block is related to allies themselves.

- **enabled –** turns ALL features of ally mod on or off. **true** if you want to play with allies and plane loadouts, **false** if you don't. Still loads crash reporter.
- **ally_bombers –** allows ships equipped with bombs to bomb only landed targets. **true** to enable, **false** to disable.
    - Bombing AI disables Zenith during bomb run and also drops flares to distract incoming Zeniths.
- **ally_delay_seconds –** how long the game waits in seconds before spawning a new ally in combat.
- **flagship_leaves –** governs ally AI behavior after player controlled ship retreats. Default is **handover,** after you retreat game automatically grants you control of an ally ship. **retreat** forces the remaining allies to pull out as well
- **retreat_key –** button used to tell allies to retreat. Uses value from Keycode table (119 corresponds to F8)
- **max_allies –** modifies how many allies can spawn at once. Theoretically there is no limit, vanilla is 2, "sane" limit is 10 because the game itself has troubles handling large fleets. Do not set this to 0, it is clamped to 1

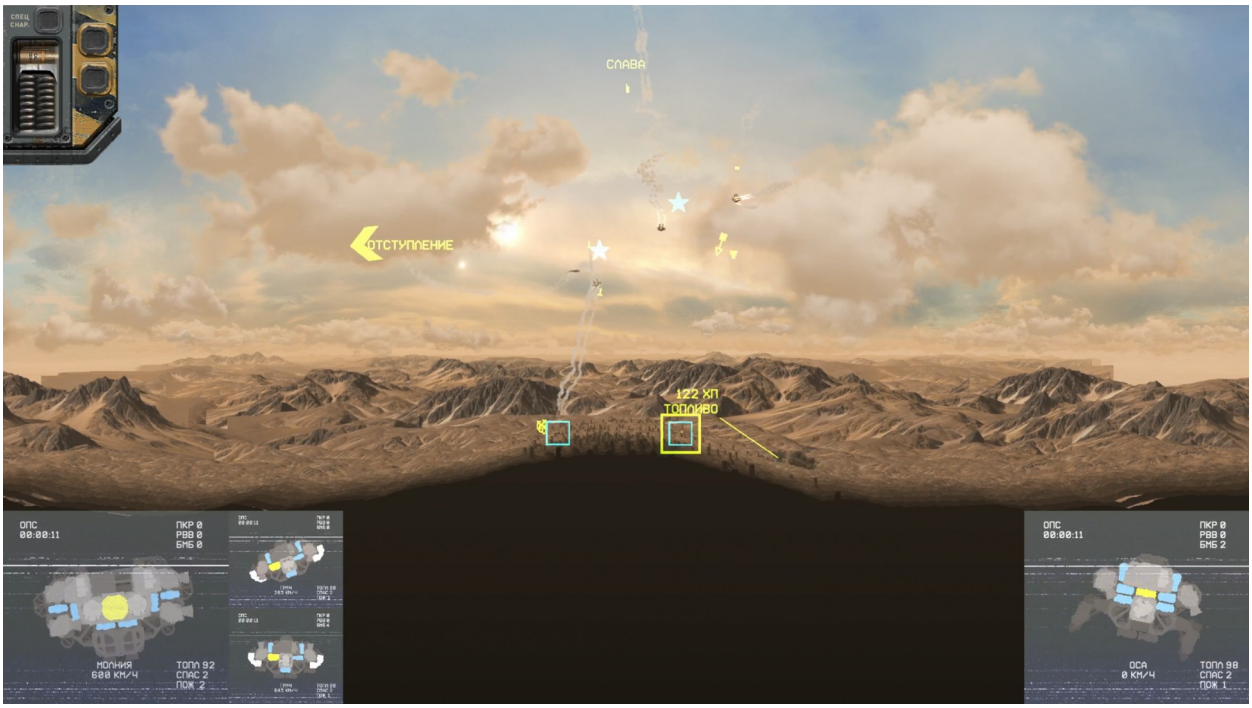
- **ship_switching –** allows to switch between ships mid-combat. TAB cycles ships. 1 switches to ship you see in top subpanel, 2 switches to ship in bottom subpanel.



Screenshot from recent mod version with debug overlay enabled. Pressing 1 switches me to Lightning, pressing 2 switches me to Fenek

- **spare_prizes –** prevents allies from treating prize ship as enemies. Leave as true, set to false if you're into attacking merchant vessels.
- **ally_zenith_autofire –** allows allies to use stock AI to fire Zeniths. Set to **false** if you feel like they are wasting their missiles and you'd rather fire them yourself after switching.
- **ally_ai_ammo –** patches default game ammo picker subroutine into allies. Admittedly it's a very lazy implementation, but it works. Set to **false** if you want allies to stick to HE shells only.
- **ally_damage_color –** modifies the color of ally star indicator and ally target reticle. Accepts both HEX and old DEC values. Game's default lime color is **A8FC00.** The provided ally config sets it to **00E4FF,** which is also the blue used by plane icons.
- **ally_panels –** renders two module view panels next to your ship. Max is two. Using more allies does not break it, but it's not fully tested behavior. Set to **false** if you find the panels distracting.

- **ally_reticle_scale –** adjusts allies' reticle size so that it doesn't overlap with your own. Set to 0 to disable. Accepts any value that is not negative.



Both Rook and I target the same ship. Because the ally reticle has different size it does not overlap my own. Another reticle can be seen on the left.

- **ally_sounds –** enables SFX for allies (enter combat and ship loss). Currently uses lines from game's folder Media/Snd/RADIO (or RADIO_ENG). I am planning to improve the voicelines in the future. Set to **false** to disable.
- **ally_fuel_retreat_percent –** allows allies to retreat once their fuel is low enough. Clamped between 0 and 100. Set to 0 to disable.
- **ally_honor_share –** crutch fix to allow all participating ships to level up. Your player-controlled ship gets 100% of honor; participating ones present on screen during combat earn 0.5 of that. 0 to disable, 1 to grant every ship more honor.

Next we have plane configuration block.

```
"plane_loadouts": true,
"extra_gunpod": true,
"plane_k13": true,
"plane_attack_delay": 1.0,
"plane_aa_range": 6000.0,
"plane_aa_delay": 1.0,
"plane_aa_angle": 0.35,
"plane_aa_speed": 1.5,
```

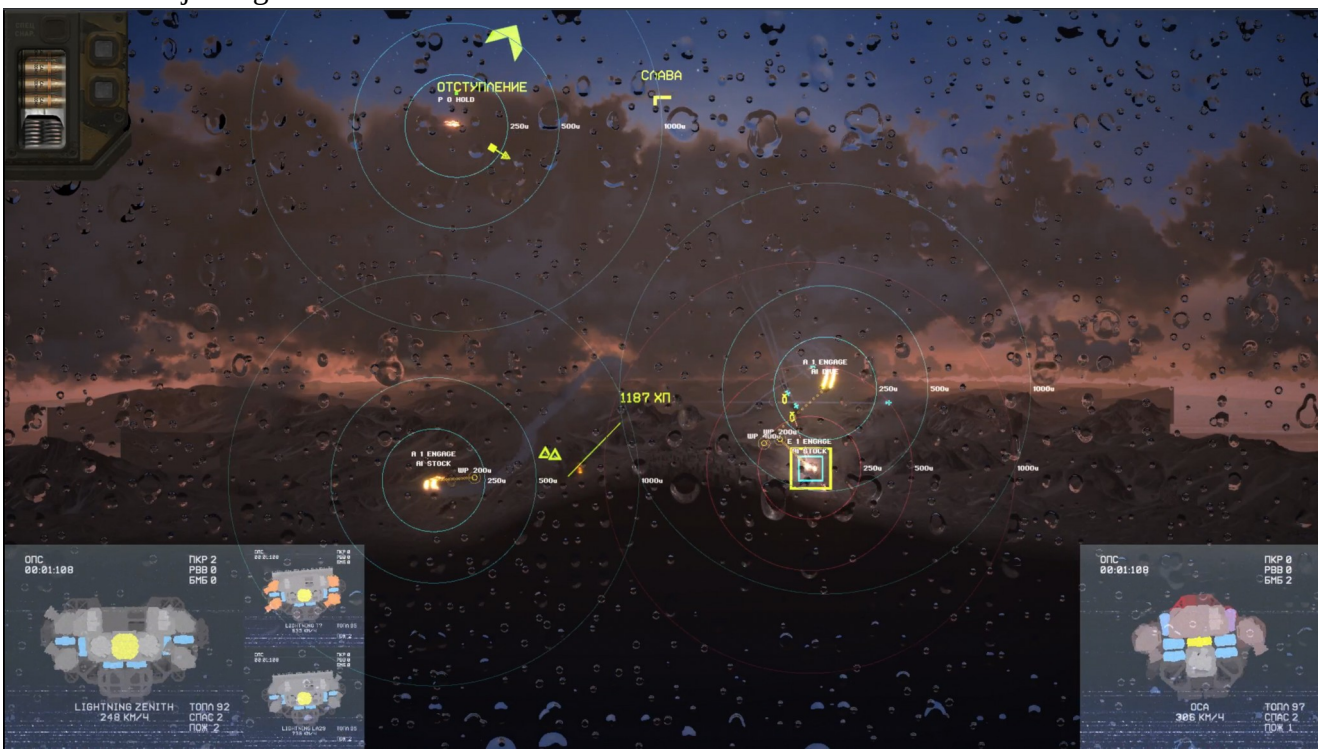


- **plane_loadouts –** allows you to manually pick any armament for planes in tac combat. Ever wanted to BOMB your enemies mid air? Now you can. Set to **false** to revert to vanilla behavior (planes automatically pick 122mm rockets or stick to 37mm gunpod).
- **extra_gunpod –** crutch fix to give every plane a gunpod so that it stays in fight after using its main payload (like bombs or rockets).
- **plane_k13 –** allows K13 to be equipped and used in tac combat. Turns your T-7s into interceptors. Great against carriers, somewhat works against Zeniths.
- **plane_attack_delay –** adds a delay before the AI routine kicks in after take off. Do not reduce it to 0 unless you like planes ramming into the flight deck and capsizing your carriers.
- **plane_aa_range –** crutch fix to make sure K-13 interceptors don't just leave off-screen and ignore all aerial threats. Uses in-game units. Vanilla is 3750.
- **plane_aa_delay –** same as **plane_attack_delay,** but exclusively for K-13.
- **plane_aa_angle –** how aligned the interceptor must be with its target. Uses radians. Game default is 0.1745 (10 degree cone), 0.35 is mod default (20 degrees). Clamped between 0.1745 and 1.2
- **plane_aa_speed –** accelerates T-7 to give them a better chance in tac combat. 1 is vanilla, 1.5 means 50% faster (ONLY when they have a target).

Remaining keys.

```
  "logging": {
    "debug_log": false,
    "debug_overlay": false,
    "gui_inspect": false,
    "log_creatures": false
  },
  "intercept_ignore_oids": [
    "MDL_MISSILE_03"
  ]
}
```

- **enemy_group_size –** how many enemies can appear in tac combat at once. Sane limit is 10.
- **enemy_all_at_once –** removes delay for spawning multiple enemies.
- **marker_text –** text you see next to FIRST slot in pre-combat order screen.
- **logging**
  - **debug_log –** extremely verbose log, mainly used for diagnostics and development. Logs per-second function calls, writes its own log in logs folder. Set to **false** for vanilla game.
  - **debug_overlay –** draws overlay in combat. Mainly useful for debugging AI features or adjusting distances in mods.



    - **gui_inspect –** tries to log every GUI interaction you have in game. Keep **false** unless diagnosing errors.
    - **log_creatures –** tracks every spawned ship. Keep **false** unless diagnosing errors
- **intercept_ignore_oids –** bit of a crutch fixed used for compatibility with mods that add more strategic missiles (mainly Map Mode Extended). Due to peculiarities of the game's code Sprint that can be launched across the strategic map becomes a valid interception target.

TUNING THE AI SYSTEM

Head to your game's folder, then Modloader – config – **hf_ai.json.** Open it with any text editor, even Notepad will do.

```
{
  "enabled": true,
  "logging": {
    "log_panic": false
  },
  "twr_cutoff": 2.0,
  "plane_group_per_launch": 3,
  "plane_group_launch_delay_seconds": 0.5,
  "ai_missile_dump_chance": 0.4,
  "ai_missile_dump_min_delay": 0.6,
  "ai_missile_dump_max_delay": 1.5,
```

- **enabled –** one master switch for ALL AI features for ships.
- **logging –** hf_ai own's logging. Mainly used for debugging flares and evasion.
- **twr_cutoff –** hard limit on ships that get evasion and tactical routines. Uses ship builder TWR, not dynamic combat TWR.
- **plane_group_per_launch –** modifies carriers to send more planes at once if they are AI controlled.
- **plane_group_launch_delay_seconds –** adds a delay between each plane in that group to prevent collisions.
- **ai_missile_dump_chance –** a chance that after launching Zenith the ship will immediately launch another one. 0.4 – 40%. Set to 0 to disable.
- **ai_missile_dump_min_delay –** minimal delay between launching Zenith in seconds. Prevents missiles destroying each other due to collision
- **ai_missile_dump_max_delay –** maximal delay between missiles if missile dump roll returned a success



```
"ai_flares": {
  "enabled": true,
  "allies": true,
  "evade": true,

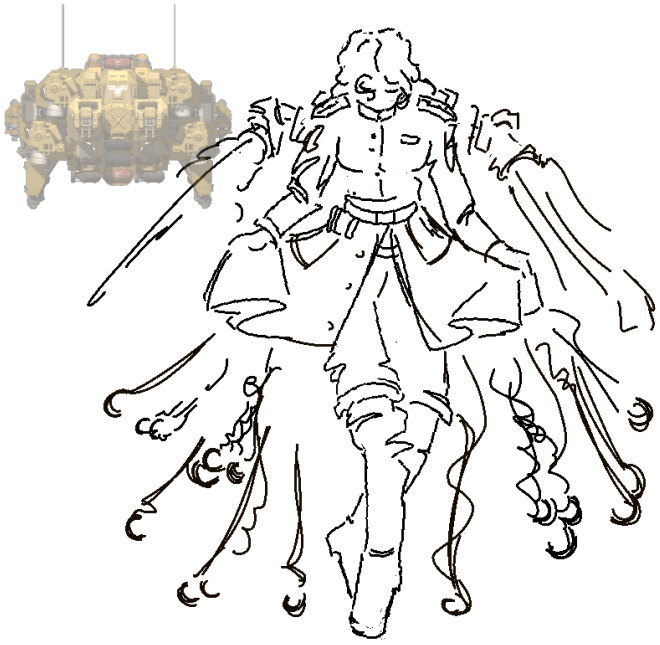
  "trigger_range": 400.0,
  "reaction_probability": 0.6,
  "burst_seconds": 1.5,

  "evasion_seconds": 3.0,
  "evasion_strength": 1.0,

  "panic_evasion_probability": 0.2,
  "panic_evasion_seconds": 3.0,
  "barrel_roll_probability": 0.001,
  "barrel_roll_seconds": 2.0,
  "panic_boost": 2.0,

  "min_interval_seconds": 1.5,

  "stray_missiles": true
},
```



- **ai_flares** block modifies the work of everything related to evading missiles and firing flares
- **enabled –** master switch for this whole block.
- **allies –** allows allied ships to also use flares and dodge missiles.
- **evade –** decides whether the ships will try to evade the incoming missile.
- **trigger_range –** range at which missile becomes a recognized threat.
- **reaction_probability –** chance for a ship to fire flares and trigger evasion.
- **burst_seconds –** how long the flare burst lasts.
- **evasion_seconds –** how long the ship will move away from missile path before returning to normal routine.
- **evasion_strength –** how much does **evasion** modify ship's built-in course. 0 effectively disables it, 1 gives the evasion routine full control.
- **barrel_roll_probability –** minor chance the enemy ship will try extremely practical maneuver of rolling away from missile.
- **barrel_roll_seconds –** duration of aforemention maneuver.
- **panic_evasion_probability –** rolled after evasion start. Same as evasion, but also uses afterburners and lasts longer.
- **panic_evasion_seconds –** duration of panic evasion in seconds.
- **panic_boost –** modifies the afterburner strength. Did you know that you can set it to 4?
- **min_interval_seconds –** how much time passes before this routine starts registering/rolling for another incoming missile.
- **stray_missiles –** attempt at making ship dodge/evade Zeniths that lost lock and just continue flying forward.



```
  },
  "tactics": {
    "enabled": true,
    "allies": true,
    "dive_enabled": true,
    "dive_under": true,
    "dive_over": true,
    "dive_chance": 0.5,
    "dive_range": 600.0,
    "dive_dwell_s": 0.1,
    "dive_cooldown_s": 1.0,
    "dive_boost": 2.0,
    "aim_enabled": true,
    "aim_chance": 0.75,
    "aim_range": 1500.0,
    "aim_tolerance": 200.0,
    "aim_boost": 2.0,
    "aim_require_fire": true,
    "aim_fire_probability": 0.5,
    "aim_fire_range": 1.6
  }
}
```

- **tactics** – a set of more nuanced rules that take into account your positioning and aim. Really makes the enemies feel more reactive in combat.
- **allies –** enables or disables the same rules for allies in combat.
- **dive_enabled –** dive prevention. Ships will try to boost away and get to one altitude with target.
- **dive_over/under –** individual toggles for underpass/overpass.
- **dive_chance –** possibility of dive evasion happening.
- **dive_range –** radius of 45 degree cone under/over the ship that detects enemies going for dive.
- **dive_dwell_s –** how long the enemy must stay in that cone to trigger the routine. At 0.1s this happens almost instantly, mod ships with 0.5s.
- **dive_cooldown_s** – how long until the check can run again.
    - target enters cone → stay in cone for **dive_dwell_s** → roll **dive_chance**
        - success: dive evasion, then **dive_cooldown_s**
        - failure: **dive_cooldown_s**, then **dwell** + roll again
- **dive_boost –** afterburner strength for pulling away. 2 is game's default
- **aim_enabled –** maneuvers that try to throw off your aim.
- **aim_chance –** chance it triggers.
- **aim_range –** range within which the enemy ship must be to trigger the aim evasion check
- **aim_tolerance –** how bad your aim can be, in units. Uses ship center, not ship total size. I should really patch that, eh?
- **aim_boost –** 2 for afterburners, 1 for default speed, 2+ for fun.
    - aim dodging routine uses some hidden values inside the code. You can modify them yourself by pasting the correct values and keys into config.
- **aim_require_fire –** the check will only happen if the enemy is firing at that moment.

p.s. qol mod packed with this mod has features that aren’t released on main yet.

1. Refuel rate adjustment: self-explanatory

2. Speed adjustment: set to classic if you want 1.151 speed formula. Or set to custom and write your own.